01 SEED Seed Initialization

Start with rare, topology-rich entities across domains



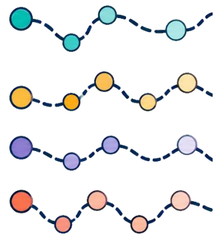
Rare-entity pool

Chinese + English Wikipedia



Topology probe

multi-start random walks



Coverage test

deep · wide · locally diverse



Domain-balanced seeds

prior labels + incremental update

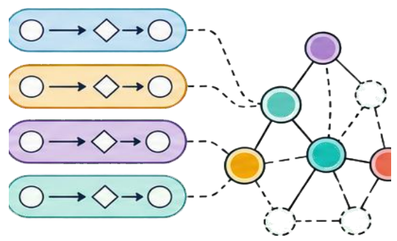
02 WEB Web-enhanced Agentic Expansion

Search, extract, locate gaps and iteratively expand



Agentic web queries

ask beyond internal knowledge



Long-tail fact extraction

web pages → relation triples



Missing-link expansion

locate gaps · query again

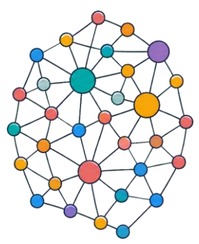


Global graph fusion

disambiguate · aggregate · iterate

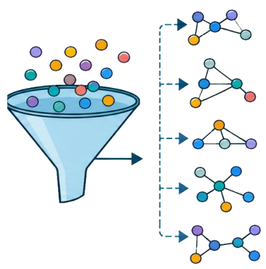
03 SAMPLE Knowledge Sampling + Difficulty Ranking

Generate variants, sample sub-graphs, rank difficulty



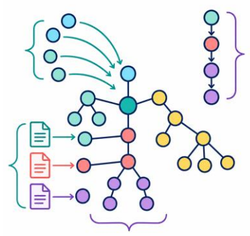
Bulk subgraph sampling

nodes · depth · width · branching



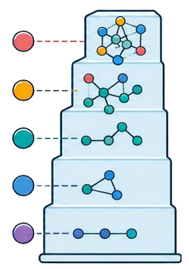
Candidate subgraphs

large-batch structural variants



Difficulty factors

degree · depth · breadth · source



Difficulty ranking

custom composite score

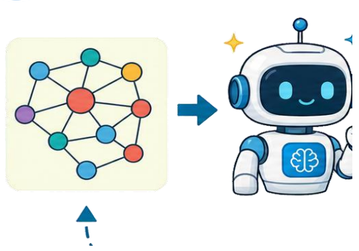


Retain hard subgraphs

high-complexity QA context

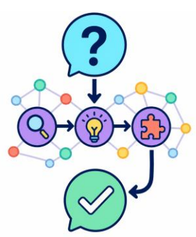
04 SYNTHESIZE QA Generation + Complexity Scaling

Scale reasoning depth, tool use, trajectory length, and style



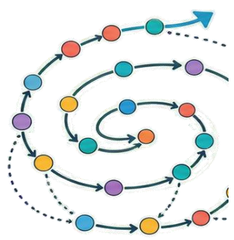
Strong-LLM synthesis

hard subgraph as context



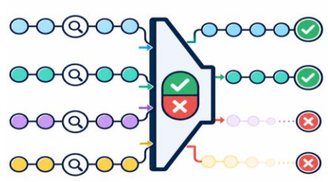
Initial QA

Q → factual evidence chain → A



Complexity scaling

more hops · deep / wide search



Pass@K Tool-use Filter

trajectory → keep ✓ / reject ×



Multi-style QA

multi-hop · wide · long-form

Hard, long-trajectory examples feedback into synthesis